Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 8 — Integration
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第8章 · 定积分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

The curve C has equation $y = (x - 1)(x - 5)$. In Figure 1, region R lies between C and the x-axis. A horizontal line $y = -3$ divides R into a lower region L and an upper region U.

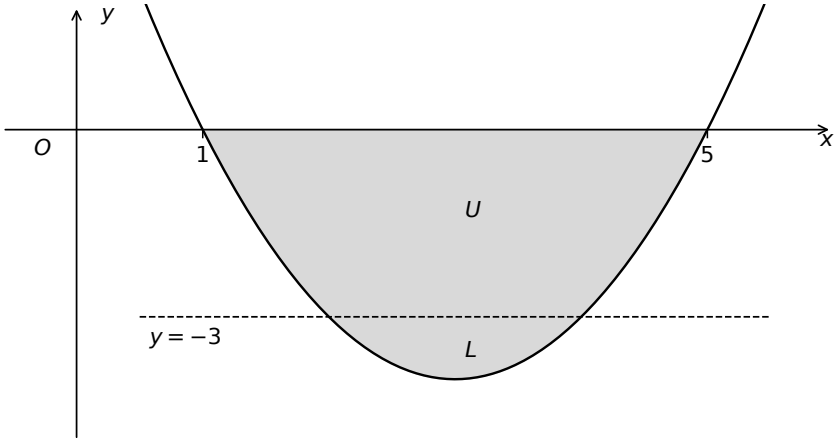


Figure 1

(a) Find where $y = -3$ meets C. (2)

(b) Find the exact area of L. (3)

(c) Find the exact area of U. (3)

(Total for Question 1 is 8 marks)

A parabolic arch above a horizontal baseline is modelled by $y = kx(6 - x)$ for $0 \leq x \leq 6$, where $k > 0$. Its cross-sectional area is 36 m^2, with x, y measured in metres. Figure 2 shows the part above the horizontal line $y = 5$.

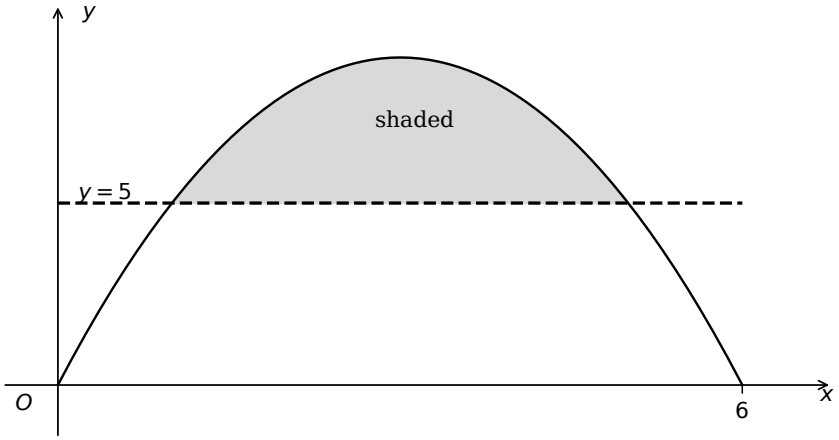


Figure 2

(a) Find k. (3)

(b) Find the width of the arch at height 5 m. (2)

(c) Find the exact fraction of the cross-sectional area above height 5 m. (3)

(Total for Question 2 is 8 marks)

Figure 3 shows the curves $y = x^3$ and $y = 4x$. They enclose two finite regions, one on each side of the y-axis.

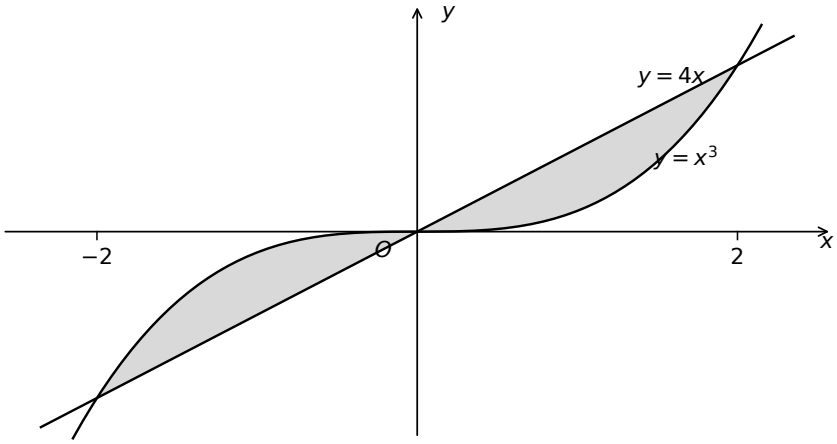


Figure 3

(a) Find all three points of intersection. (3)

(b) Find the total area of the two finite regions. (3)

(c) A student uses $\int_{-2}^{2}(4x - x^3)dx$ and obtains zero. Explain why this does not give the required total area. (2)

(Total for Question 3 is 8 marks)

The line l is tangent to $y = \sqrt{x}$ at $P(4, 2)$. Figure 4 shows the finite region bounded by l, the curve and the y-axis.

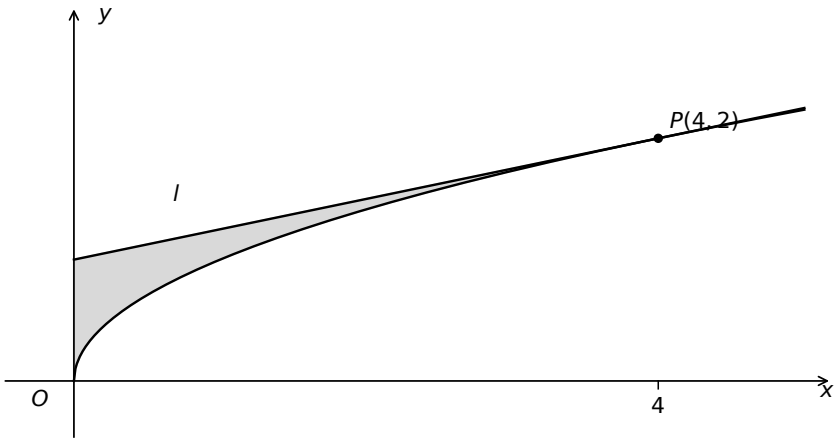


Figure 4

(a) Find the equation of l. (2)

(b) Find the exact shaded area. (3)

(c) Without calculating a new integral, find the area bounded by $y = \sqrt{x} + 3$, its tangent at $x = 4$, and the y-axis. Give a reason. (2)

(Total for Question 4 is 7 marks)

Figure 5 shows $y = \sqrt{x}$, the line $y = x - 2$, and the x-axis. The shaded finite region is bounded by all three.

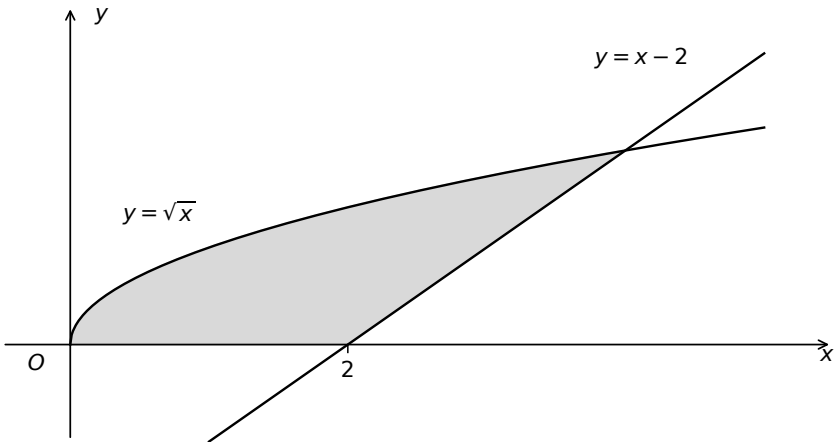


Figure 5

(a) Find the intersection of the curve and line, checking the sign condition before squaring. (3)

(b) Find the exact shaded area. (3)

(c) Explain why $\int_0^4 [\sqrt{x} - (x - 2)]dx$ gives too large a result. (2)

(Total for Question 5 is 8 marks)

For $x \geq 0$, $f(x) = 3x^2 - 12x + 9$. A positive number b satisfies $\int_0^b f(x)dx = 0$.

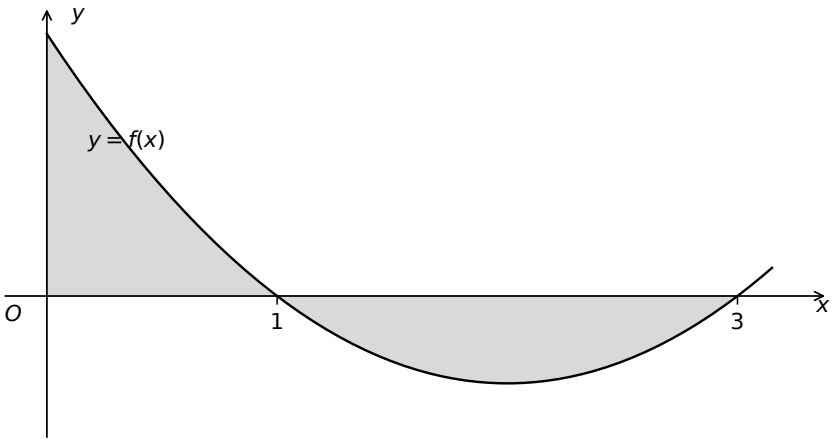


Figure 6

(a) Find b, showing all algebraic steps. (3)

(b) Find the total area between the graph and the x-axis on $0 \leq x \leq b$. (3)

(c) Explain how your answers to (a) and (b) can both be correct. (2)

(Total for Question 6 is 8 marks)

For $0 \leq x \leq 2$, $f(x) = x^2 + k$, where $k > 0$. Let $I = \int_0^2 f(x)dx$ and let T be its estimate using two equal trapezia.

(a) Find I and T in terms of k, and hence show that $T - I$ is independent of k. (3)

(b) The relative error is defined as $(T - I)/I$.

(i) Find k if this relative error is 5%. (2)

(ii) determine whether the relative error increases or decreases as k increases, giving a reason. (2)

(Total for Question 7 is 7 marks)

The table gives values of a continuous function.

x	0	1	2	3	4
$f(x)$	0	2	h	5	0

The trapezium estimate of $\int_0^4 f(x)dx$ using four equal strips is 12.

(a) Find h. (2)

(b) Using only $x = 0, 2, 4$, find a second trapezium estimate. (2)

(c) A student claims that the two estimates prove $10 \leq \int_0^4 f(x)\,dx \leq 12$. State whether this is justified by the table alone, giving one reason. (2)

(Total for Question 8 is 6 marks)

A hall roof has cross-section $y = k(9 - x^2)$ for $-3 \leq x \leq 3$, where x, y are in metres. Its greatest height above the baseline is 2.7 m. The hall has constant cross-section along its 20 m length. Figure 9 shows the roof and the horizontal level $y = 1.2$.

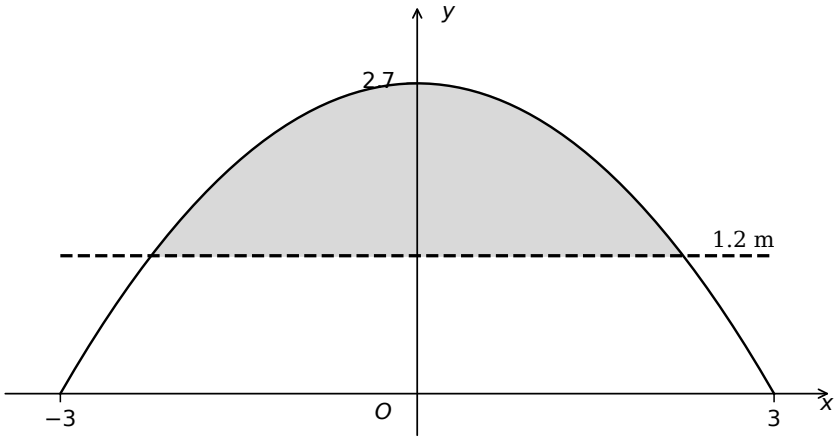


Figure 9

(a) Find k and the cross-sectional area below the roof and above the baseline. (3)

(b) Find the volume of the hall below the roof and above the baseline. (2)

(c) Find the exact cross-sectional area above $y = 1.2$. (3)

(Total for Question 9 is 8 marks)

A channel cross-section has depths, in metres, measured below a horizontal water surface:

x **(m)**	0	2	4	6	8
depth	0	0.8	1.6	1.4	0

For this question, model the bed by straight line segments joining consecutive measurement points. The water surface is then lowered by 0.2 m.

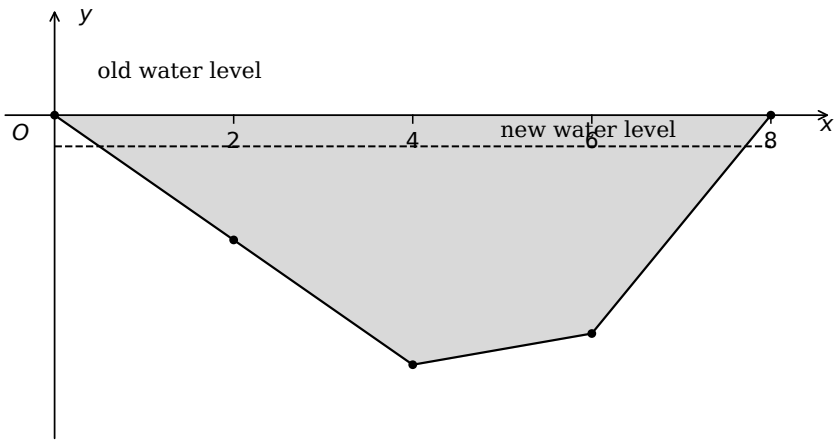


Figure 10

(a) Find the original cross-sectional water area. (2)

(b) The water level is lowered by 0.2 m, while the piecewise-linear bed model remains unchanged.

(i) Find the new positions of the left and right water edges. (3)

(ii) Find the new cross-sectional area. Explain why simply subtracting $8(0.2)$ from the old area is not exact. (4)

(Total for Question 10 is 9 marks)

Let $I = \int_1^4 \sqrt{x}\, dx$.

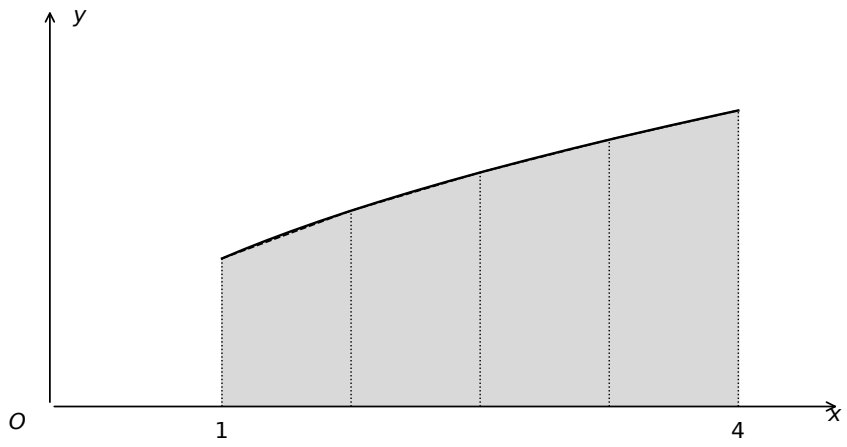


Figure 11

(a) Use four equal trapezia to estimate I, giving the result to 4 decimal places. Keep unrounded function values in your calculation. (3)

(b) Find I exactly and hence the absolute error of your estimate, to 3 significant figures. (3)

(c) State whether the trapezium estimate in part (a) is an overestimate or an underestimate of the integral. Give a reason based on the position of each chord relative to the curve. (2)

(Total for Question 11 is 8 marks)

The curve $y = x^2 - 4$ lies at or below the x-axis on $0 \leq x \leq 2$. Figure 12 shows two equal trapezia between this curve and the axis.

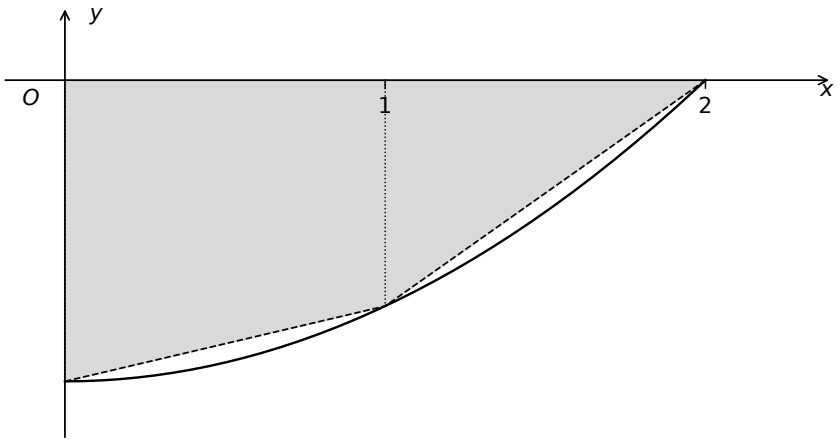


Figure 12

(a) Use these two trapezia to estimate the positive area between the curve and the axis. (3)

(b) Find the exact area and determine whether the estimate is too large or too small. (3)

(c) For the signed integral $\int_0^2 (x^2 - 4)dx$, is the trapezium estimate an overestimate or an underestimate? Give numerical evidence. (2)

(Total for Question 12 is 8 marks)

The horizontal line $y = h$, where $h > 0$, and the curve $y = x^2$ enclose a finite region R, as shown in Figure 13.

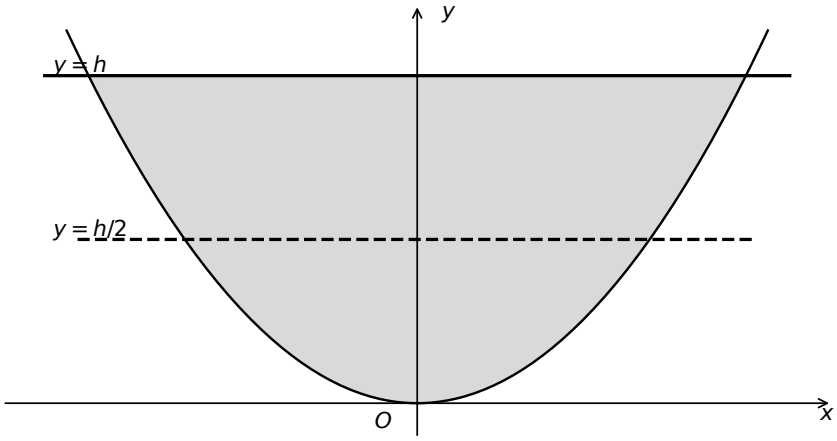


Figure 13

(a) Show that the area of R is $\frac{4}{3}h^{3/2}$. (3)

(b) The area of R is 36. Find h and the two points of intersection. (2)

(c) Find the fraction of R lying above $y = h/2$. (3)

(Total for Question 13 is 8 marks)

The upper boundary of a moulded strip is modelled by $y = ax^2 + b$ for $0 \le x \le 2$. Lengths are in millimetres. The cross-section has vertical sides at $x = 0, 2$ and base $y = 0$. At the right edge it is 5 mm thick, and its cross-sectional area is $6\ \text{mm}^2$.

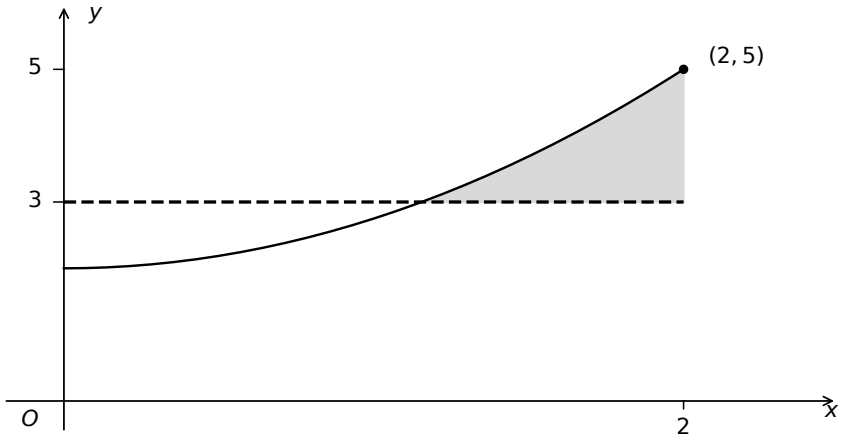


Figure 14

(a) Form two equations in a, b and find them. (3)

(b) Find the position at which the strip first reaches height 3 mm. (2)

(c) Find the exact cross-sectional area of material lying above the level $y = 3$. (3)

(Total for Question 14 is 8 marks)

For $1 \leq x \leq 4$, the curve C has equation $y = (x^2 + 4)/\sqrt{x}$. The chord joins its endpoints at $x = 1$ and $x = 4$. Figure 15 shows that the chord lies above the curve between these endpoints.

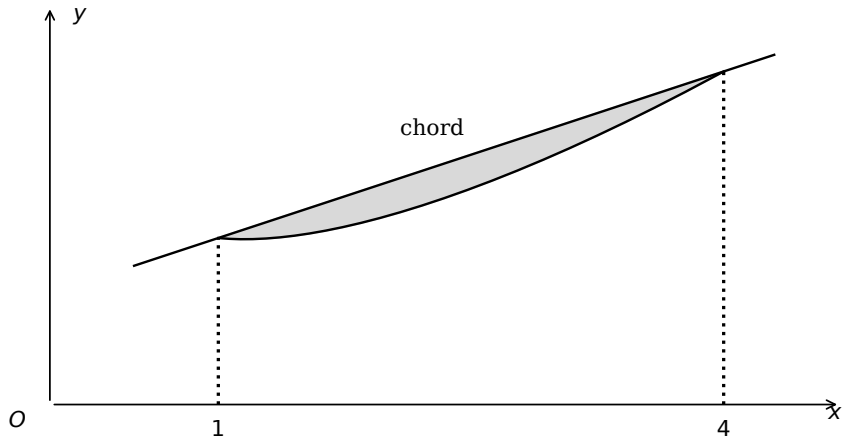


Figure 15

(a) Find the equation of the chord. (3)

(b) Find $\int_1^4 y\,dx$ exactly. (3)

(c) Find the exact area between the chord and curve. (2)

(Total for Question 15 is 8 marks)

The curve $y = 2^x$ on $0 \leq x \leq 3$ is shown in Figure 16 inside the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 8$.

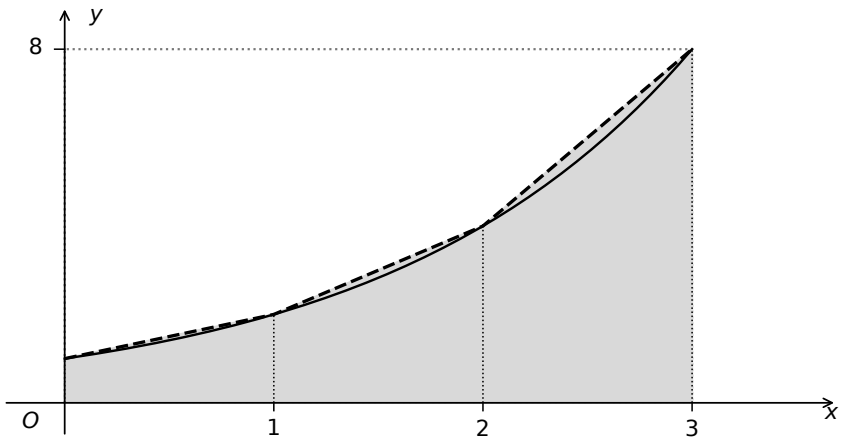


Figure 16

(a) Use three equal trapezia to estimate the area below the curve and above the x-axis. (3)

(b) Hence estimate the area inside the rectangle but above the curve. (2)

(c) The curve lies below each chord joining consecutive tabulated points. State whether your answer to (b) overestimates or underestimates the true area. (2)

(Total for Question 16 is 7 marks)

The boundary of a panel is modelled by $f(x) = (x + 2)^2$ for $-2 \leq x \leq 0$, and $f(x) = 4 - 2x$ for $0 < x \leq 2$. Its base is the x-axis. A vertical line $x = a$, where $0 < a < 2$, divides the panel into two equal areas.

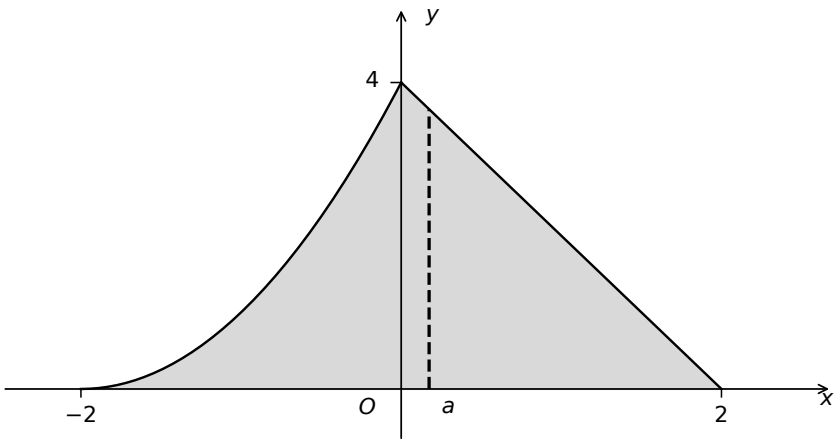


Figure 17

(a) Find the total panel area. (3)

(b) Show that $a^2 - 4a + 2/3 = 0$. (3)

(c) Find a exactly, explaining your choice of root. (3)

(Total for Question 17 is 9 marks)

The line $y = 2x + 3$ and curve $y = x^2$ enclose a finite region. The y-axis divides it into regions L and R, as shown in Figure 18.

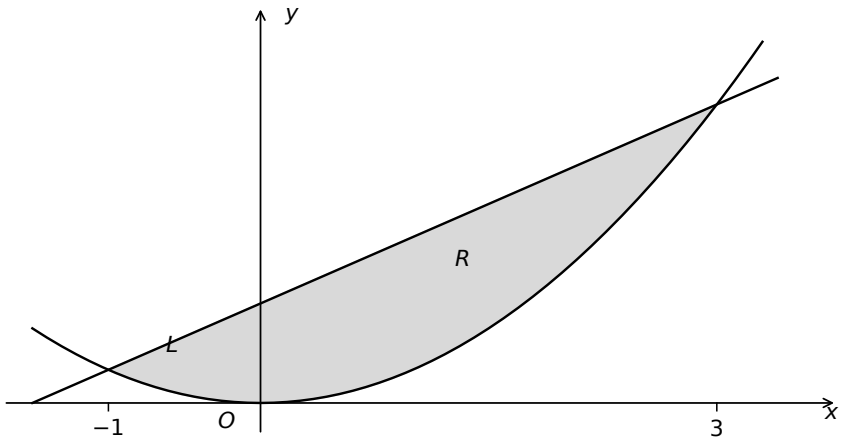


Figure 18

(a) Find their two points of intersection. (2)

(b) Find the exact area of L. (3)

(c) Find the ratio of the areas $L : R$ in simplest integer form. (2)

(Total for Question 18 is 7 marks)

Figure 19 shows a continuous curve $y = f(x)$ on $[-2, 4]$. It is above the x-axis for $-2 < x < 0$ and below it for $0 < x < 4$. The two shaded areas are 7 and 3 square units, respectively. No formula for f is given.

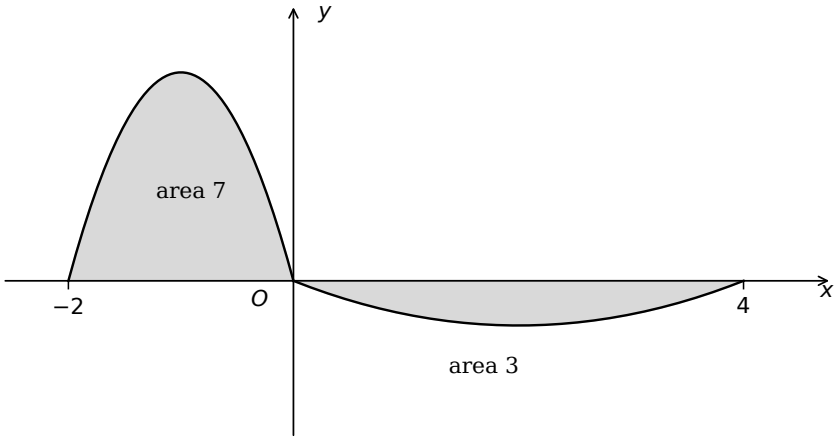


Figure 19

(a) Find $\int_{-2}^{4} f(x)dx$ and $\int_{0}^{4} f(x)dx$. (2)

(b) Using a horizontal scale factor, find $\int_{-1}^{2} f(2x)dx$. Do not use an unknown antiderivative. (3)

(c) Find $\int_{-4}^{2} f(-x)dx$, explaining the graph transformation. (2)

(Total for Question 19 is 7 marks)

A panel has a straight base of length 6 m. Perpendicular heights of its top boundary are measured as follows:

distance x (m)	0	1	3	6
height y (m)	1	2	4	3

For a model, join consecutive measured points by straight lines.

(a) State why the widths of the trapezia must be treated separately for these data. (2)

(b) Find the modelled panel area. (3)

(c) Both faces require 0.4 litres of coating per square metre per face. Coating is sold in 5-litre tins. Find the minimum number of tins required under the model. (2)

(Total for Question 20 is 7 marks)

Let $f_k(x) = x^2 - 2kx + k$ on $0 \le x \le 1$, where k is real. Figure 21 shows several members of this family.

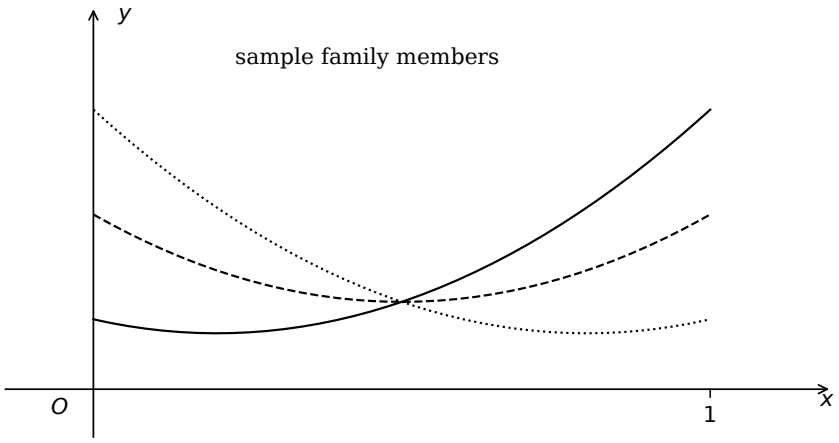


Figure 21

(a) Show that $\int_0^1 f_k(x)dx = 1/3$ for every k. (3)

(b) Find the complete range of k for which $f_k(x) \ge 0$ throughout $[0, 1]$. (3)

(c) Using one trapezium, find the area estimate when $0 \le k \le 1$, and its absolute error. Explain why both are independent of k. (3)

(Total for Question 21 is 9 marks)

The curve C has equation $y = x^2 - 4x + 5$. The line l is tangent to C at the point with $x = 3$. The shaded region in Figure 22 lies between C, l and the lines $x = 1, x = 5$.

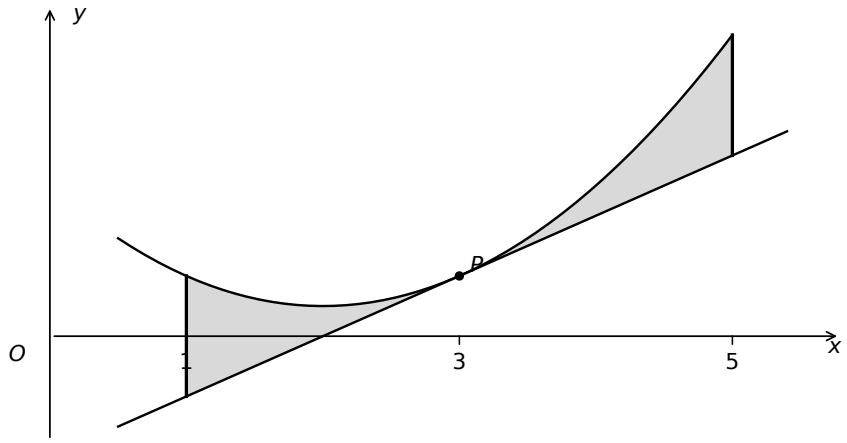


Figure 22

(a) Find l and show that the vertical separation of C above l is $(x - 3)^2$. (4)

(b) Find the total shaded area exactly. (3)

(c) Without evaluating another integral, explain why $x = 3$ bisects the shaded area. (2)

(Total for Question 22 is 9 marks)

Figure 23 gives a continuous curve $y = f(x)$ with $f(x) \geq 0$ on $[0, 2]$ and $f(x) \leq 0$ on $[2, 5]$. It is known that $\int_0^2 f(x)dx = 5$ and $\int_0^5 f(x)dx = -1$. Define $g(x) = f(x) + k$.

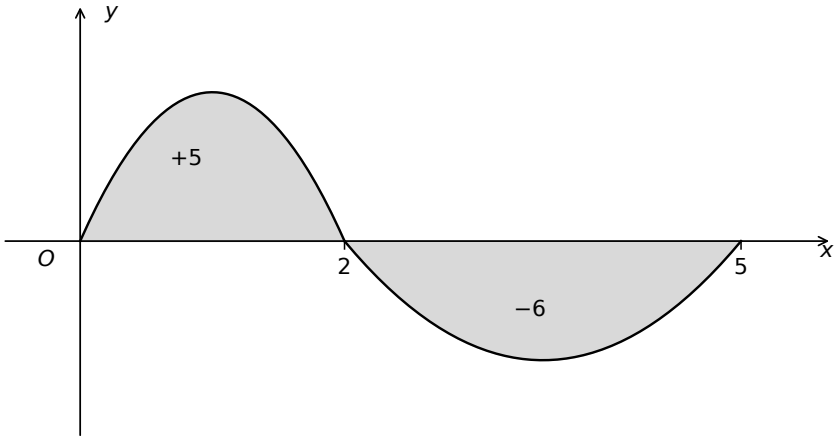


Figure 23

(a) Find the positive area between f and the x-axis on $[2, 5]$. (2)

(b) Find k if $\int_2^5 g(x)dx = 0$. (3)

(c) For the value of k found in part (b), explain why $\int_2^5 g(x)\,dx = 0$ does not determine the total positive area between the graph of g and the x-axis. (2)

(Total for Question 23 is 7 marks)

A planting area is bounded by $y = 4$, $y = 1 + \sqrt{x}$ and the y-axis, as shown in Figure 24. Coordinates are in metres. A project is viable only if the planting area is at least $9.5\ \text{m}^2$.

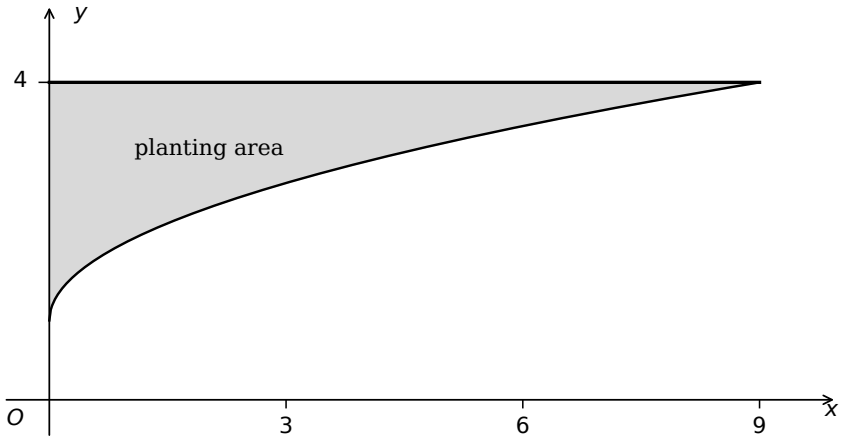


Figure 24

(a) Using the points $x = 0, 3, 6, 9$ on the curved boundary, obtain a trapezium-based estimate of the planting area. (4)

(b) Use integration to find the exact planting area. (3)

(c) State the correct project decision and explain why using the estimate alone could lead to the wrong decision. (2)

(Total for Question 24 is 9 marks)

On $0 \leq x \leq 4$, two profiles are $y = x^2$ and $y = k\sqrt{x}$, with $k > 0$. The areas beneath the two profiles over this interval are equal. Figure 25 shows the profiles after k has been determined.

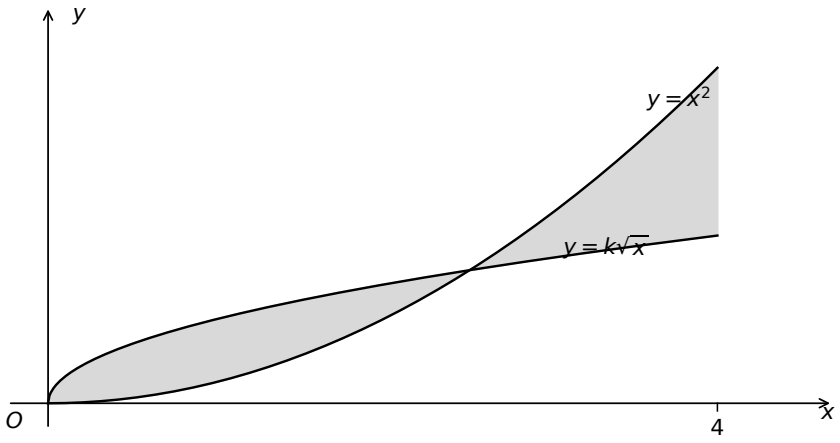


Figure 25

(a) Find k. (3)

(b) Find the positive x-coordinate at which the profiles cross. (2)

(c) Find the total area between the profiles on $[0, 4]$. (4)

(Total for Question 25 is 9 marks)

The curve C is $y = x(4 - x)$ for $0 \leq x \leq 4$. A horizontal line at height h, where $0 < h < 4$, cuts off a region between the line and the top of the curve.

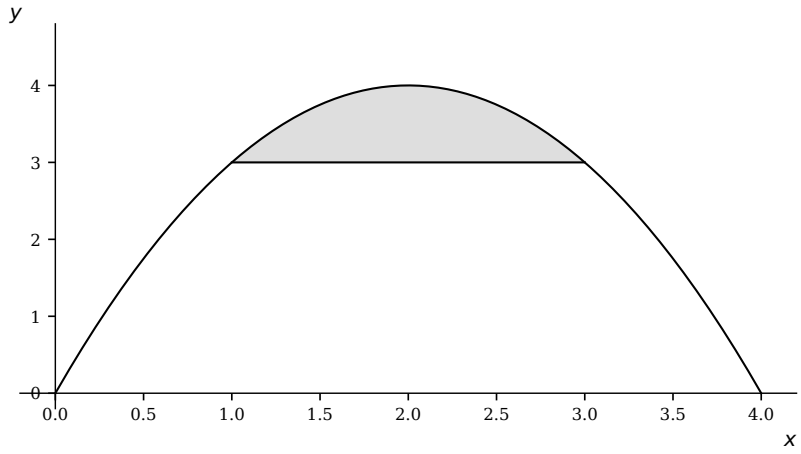


Figure 26

(a) Show that its intersection coordinates have $x = 2 \pm \sqrt{4 - h}$. (3)

(b) Given that the chord has length 2, find

(i) the value of h, (2)

(ii) the exact area above the chord and below C. (3)

(Total for Question 26 is 8 marks)

The table gives values of a continuous positive function f on $[0, 4]$.

x	0	1	2	3	4
$f(x)$	2.0	2.8	3.4	3.8	4.0

Write $I = \int_0^4 f(x)dx$.

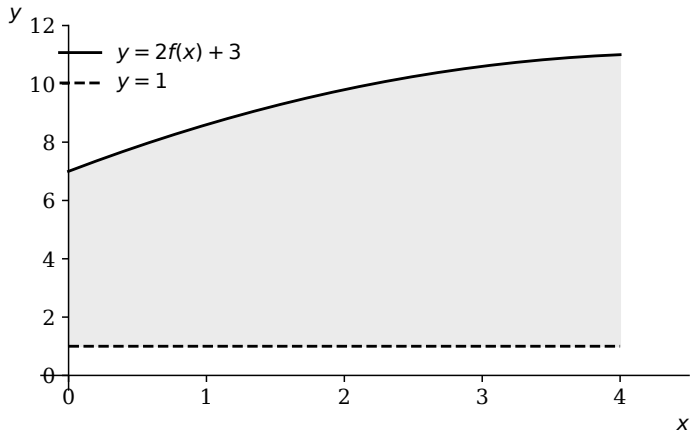


Figure 27

(a) Use the trapezium rule with all the tabulated values to estimate I. (2)

(b) For $0 \leq x \leq 4$, the upper curve is $y = 2f(x) + 3$ and the lower line is $y = 1$.

(i) Use your answer to (a) to estimate the area between these graphs. (2)

(ii) The graph of f lies above each chord joining consecutive tabulated points. Determine whether the estimate in (b)(i) is an overestimate or an underestimate, giving a reason. (2)

(Total for Question 27 is 6 marks)

A positive curve $y = f(x)$ is defined for $0 \leq x \leq 6$. A trapezium-rule estimate using equally spaced ordinates gives $\int_0^6 f(x)dx \approx 18.4$. In the rectangle $0 \leq x \leq 6$, $0 \leq y \leq 5$, the curve divides the rectangle into two regions. The curve lies entirely inside the rectangle.

(a) Estimate the ratio of the area below the curve to the area above the curve, in simplest integer form. (2)

(b) The curve lies below each straight chord used by the trapezium rule.

(i) State whether the estimated area above the curve is too large or too small. (2)

(ii) Determine whether the ratio in (a) is too large or too small, explaining your answer. (2)

(Total for Question 28 is 6 marks)

The curve C has equation $y = x^2 - 4x + 6$. Its tangent at $x = 1$ is l. The finite region R is bounded by C, l and the lines $x = 0$ and $x = 3$.

(a) Find an equation for l and show that the vertical separation between C and l is $(x - 1)^2$. (3)

(b) Find

(i) the exact area of R. (2)

(ii) the value of a, where $0 < a < 3$, such that $x = a$ divides R into two equal areas. (3)

(Total for Question 29 is 8 marks)

The curve $y = f(x)$ passes through $(1, 0)$ and has $f\prime(x) = 4x - 6$.

(a) Find $f(x)$ and the other point where the curve crosses the x-axis. (3)

(b) Let A be the total area between the curve and the x-axis on $0 \leq x \leq 2$.

(i) Find A exactly. (2)

(ii) Find $\int_0^2 f(x)dx$ and explain why this differs from A. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C08-R01 · 8 marks

(a) · 2 marks

$(2, -3)$ and $(4, -3)$.

联立$x^2-6x+5=-3$，得$(x-2)(x-4)=0$。

M1	联立并解二次式
A1	两交点

(b) · 3 marks

$4/3$ square units.

下方区域位于x=2至4，面积 $\int_2^4[-3-(x^2-6x+5)]dx=[-x^3/3+3x^2-8x]_2^4=4/3$。

M1	正确上下函数及积分界
M1	正确原函数
A1	精确面积

(c) · 3 marks

$28/3$ square units.

总区域在x=1至5且曲线在横轴下，面积 $-\int_1^5(x^2-6x+5)dx=32/3$；扣下方4/3得28/3。

M1	正确求完整负值区的面积
M1	减去L
A1	精确面积

Question 2

P2-C08-R02 · 8 marks

(a) · 3 marks

$k = 1$.

横截面积 $\int_0^6 k(6x - x^2)dx = 36k$，与36相等给k=1。

M1	用面积建立定积分方程
M1	正确积分及代界
A1	k

(b) · 2 marks

4 m.

解$6x-x^2=5$得x=1、5，因此宽度4，不是任一交点横坐标。

M1	解水平截线交点
A1	宽度

(c) · 3 marks

$8/27$.

上帽形面积 $\int_1^5 (6x - x^2 - 5)dx = 32/3$；除以总面积36得$8/27$。

M1	正确帽形积分
A1	帽形面积
A1	占原总面积的比例

Question 3

P2-C08-R03 · 8 marks

(a) · 3 marks

$(-2,-8),(0,0),(2,8)$.

联立得x(x−2)(x+2)=0，再代直线求纵坐标。

M1	联立并因式分解
A1	三个横坐标
A1	三交点

(b) · 3 marks

8 square units.

在0到2直线在上，面积 $\int_0^2 (4x - x^3)dx = 4$；左侧反转上下函数但面积同为4，总8。

M1	分清两区上下关系或用对称性
M1	正确积分
A1	总面积

(c) · 2 marks

The integrand changes sign at 0, so equal positive and negative signed areas cancel.

左侧$4x-x^3$为负，右侧为正；积分相抵不表示几何面积为0，必须分区后取正。

B1	指出符号改变
B1	有向面积抵消与面积的区别

Question 4

P2-C08-R04 · 7 marks

(a) · 2 marks

$y = x/4 + 1$.

导数$1/(2\sqrt{x})$在4为$1/4$，代点P得$y-2=(x-4)/4$。

M1	求导及切点斜率
A1	切线方程

(b) · 3 marks

$2/3$ square units.

0到4内直线在上，面积 $\int_0^4 (x/4 + 1 - x^{1/2})dx = [x^2/8 + x - 2x^{3/2}/3]_0^4 = 2/3$。

M1	设置上减下与积分界
M1	正确原函数
A1	精确面积

(c) · 2 marks

$2/3$: both boundaries are translated vertically by 3, so their separation is unchanged.

同时平移曲线和相应切线不会改变两者的纵向差及横向宽度，故面积保持不变。

B1	平移后面积正确
B1	两边界一起平移的理由

Question 5

P2-C08-R05 · 8 marks

(a) · 3 marks

$(4, 2)$.

√x=x−2要求x≥2。平方给$x^2-5x+4=0$，候选1和4；1使原式右边为负，故舍弃。

M1	写出平方前的符号限制
M1	解二次式
A1	只保留有效交点

(b) · 3 marks

$10/3$ square units.

曲线下0到4面积16/3，减去直线与横轴之间x=2到4的三角形面积2，得10/3。若分段积分，0至2下边界是横轴。

M1	正确分界x=2或减去三角形
M1	正确积分曲线
A1	面积

(c) · 2 marks

For $0 < x < 2$ the line is below the x-axis; that integral includes an extra triangle of area 2.

所求阴影在0至2以下边界为横轴，不是线y=x−2；错误积分额外包含横轴以下底2高2的三角形。

B1	识别错误区间和下边界
B1	额外面积2

Question 6

P2-C08-R06 · 8 marks

(a) · 3 marks

$b = 3$.

积分得$b^3-6b^2+9b=b(b-3)^2=0$；b>0排除0，故3。

M1	正确积分并代未知上限
M1	因式分解并解方程
A1	正上限

(b) · 3 marks

8 square units.

f=3(x−1)(x−3)，在0至1正、1至3负。原函数$F=x^3-6x^2+9x$，F(1)=4、F(0)=F(3)=0，两面积各4，总8。

M1	找到符号切换点1
M1	按符号分两积分
A1	两正面积和8

(c) · 2 marks

The signed integral is $4 - 4 = 0$, while total geometric area is $4 + 4 = 8$.

定积分计正负方向，几何面积取正；两个相等的面积可在积分中抵消。

B1	正负积分关系
B1	面积相加关系

Question 7

P2-C08-R07 · 7 marks

(a) · 3 marks

$I = 8/3 + 2k$, $T = 3 + 2k$, $T - I = 1/3$.

积分得$I = 8/3 + 2k$。表值为$k, 1 + k, 4 + k$，步长1，所以$T = \frac{1}{2}[k + (4 + k) + 2(1 + k)] = 3 + 2k$；相减为$1/3$，参数消去。

M1	精确积分
M1	两梯形估计
A1	误差常数

(b) The relative error is defined as $(T - I)/I$.

(i) · 2 marks

$k = 2$.

$(1/3)/(8/3 + 2k) = 1/20$，得$I = 20/3$，所以$k = 2$。

M1	正确相对误差方程
A1	参数

(ii) · 2 marks

Decreases.

绝对误差固定为正数$1/3$，而精确积分$8/3 + 2k$随k增加而增大，所以比值变小。无需用商法则。

B1	固定误差与增加分母
B1	正确方向

Question 8

P2-C08-R08 · 6 marks

(a) · 2 marks

$h = 5$.

步长1，估计½[0+0+2(2+h+5)]=h+7=12，故h=5。

M1	正确权重与步长
A1	h

(b) · 2 marks

10.

此时步长2，估计2/2[0+0+2×5]=10；不能继续用第一问的步长1。

M1	正确较粗步长与节点
A1	估计值

(c) · 2 marks

No. Values or shape between the tabulated points are not specified, so neither estimate is known to be an upper or lower bound.

连续不意味着曲线必在每条连接线同一侧；这些采样点之间可能向上或向下弯，不能把两种近似当作一对界。

B1	不能下此结论
B1	指出缺少曲线形状或界的依据

Question 9

P2-C08-R09 · 8 marks

(a) · 3 marks

$k = 3/10$; area $10.8\ \mathrm{m}^2$.

顶点x=0给9k=2.7；面积 $\int_{-3}^{3}(2.7 - 0.3x^2)dx = 10.8$。

M1	用最高点定k
M1	正确积分界与表达式
A1	横截面积

(b) · 2 marks

$216\ \mathrm{m}^3$.

截面恒定，体积=横截面积×长度=10.8×20；此处不使用旋转体积分。

M1	恒截面棱柱关系
A1	体积及单位

(c) · 3 marks

$2\sqrt{5}\ \mathrm{m}^2$.

截线满足$2.7-0.3x^2=1.2$，故x=$\pm\sqrt{5}$。面积 $\int_{-\sqrt{5}}^{\sqrt{5}}(1.5 - 0.3x^2)dx = 2\sqrt{5}$。

M1	求截线左右交点
M1	正确上减下的积分
A1	精确面积

Question 10

P2-C08-R10 · 9 marks

(a) · 2 marks

$7.6\ \mathrm{m}^2$.

分段直线模型下梯形求和给准确模型面积：2/2[0+0+2(0.8+1.6+1.4)]=7.6。

M1	正确步长与梯形权重
A1	原面积

(b) The water level is lowered by 0.2 m, while the piecewise-linear bed model remains unchanged.

(i) · 3 marks

$x = 1/2$ m and $x = 54/7$ m.

左岸首段深度=0.4x，达到0.2处x=0.5；右岸最后段深度=0.7(8−x)，达到0.2处x=8−2/7=54/7。

M1	用左岸线性深度关系
M1	用右岸线性深度关系
A1	两个新边界

(ii) · 4 marks

$851/140 \approx 6.08\ \mathrm{m}^2$.

初算7.6−1.6=6，把已露出的两岸也当成负水深；需加回两小三角形，面积½×0.5×0.2=0.05和½×(2/7)×0.2=1/35。总6+1/20+1/35=851/140。

M1	从旧面积减矩形或重新分段求和
M1	正确处理新干燥的两岸
A1	新面积
B1	说明固定8m宽扣法的错误

Question 11

P2-C08-R11 · 8 marks

(a) · 3 marks

$T = \frac{3}{8}[3 + 2(\sqrt{7}/2 + \sqrt{10}/2 + \sqrt{13}/2)] \approx 4.6551.$

步长(4−1)/4=3/4，内部节点7/4、5/2、13/4，函数值分别√7/2、√10/2、√13/2；端点值1和2。

B1	正确步长和内部节点
M1	正确梯形公式
A1	估计及精度

(b) · 3 marks

$I = 14/3$; error ≈ 0.0116.

原函数2x^(3/2)/3，代界得14/3。误差是|精确值−未舍入T|，不是用近似积分本身作分母。

M1	正确原函数及定积分
A1	精确积分
A1	绝对误差与精度

(c) · 2 marks

The curve lies above its chords, so the trapezia underestimate the integral.

平方根图像向下弯，每段连接线在曲线下；图中所有高度为正，因此梯形面积总和小于真实面积。

B1	曲线与弦的上下关系
B1	正确低估结论

Question 12

P2-C08-R12 · 8 marks

(a) · 3 marks

5 square units.

用正深度4、3、0求梯形和½[4+0+2×3]=5；若先求有向积分得到−5，最后再取正。

M1	正确处理负函数值为正深度
M1	两梯形权重
A1	正面积估计

(b) · 3 marks

$16/3$ square units; the estimate is too small by $1/3$.

几何面积 $\int_0^2 (4-x^2)dx = 8 - 8/3 = 16/3$，减估计5得1/3。

M1	设置正确面积积分
A1	精确面积
A1	比较及误差

(c) · 2 marks

Overestimate: $-5 > -16/3$.

有向积分越负越小；虽正面积被低估，但其负值−5反而大于真实有向积分−16/3。

B1	正确有向比较
B1	结论与数值依据

Question 13

P2-C08-R13 · 8 marks

(a) · 3 marks

$\int_{-\sqrt{h}}^{\sqrt{h}}(h-x^2)dx=\frac{4}{3}h^{3/2}$.

先由交点x=±√h确定积分界；利用对称或直接代入原函数hx−x^3/3，得到目标。

M1	以h表达两积分界
M1	正确上减下及原函数
A1	推出目标式

(b) · 2 marks

$h=9;\ (-3,9),(3,9)$.

4h^(3/2)/3=36得h^(3/2)=27，取正h=9；交点横坐标±3。

M1	解正参数的幂方程
A1	h与两点

(c) · 3 marks

$1-1/(2\sqrt{2})$.

先求截线下方：它恰好是y=x^2与y=h/2围成的同形区域，面积(4/3)(h/2)^(3/2)，占整个R的1/(2√2)。所求为截线上方，因此必须用1减去这个比例。

M1	正确识别截线下方的同形区域
M1	计算下部比例后取补集
A1	精确上部比例

Question 14

P2-C08-R14 · 8 marks

(a) · 3 marks

$4a+b=5,\ 8a/3+2b=6;\ a=3/4,b=2.$

右端点给4a+b=5；截面积为0到2的积分，得到8a/3+2b=6，联立。

M1	点条件方程
M1	积分面积方程
A1	正确参数

(b) · 2 marks

$x=2/\sqrt{3}\ \text{mm}.$

解$3x^2/4+2=3$，得$x^2=4/3$，在$[0,2]$取正根；该模型在$x\geqslant 0$单调递增。

M1	解厚度方程并限制范围
A1	有效位置

(c) · 3 marks

$4/(3\sqrt{3})\ \text{mm}^2.$

该部分仅在$x=2/\sqrt{3}$到2存在，积分 $\int_{2/\sqrt{3}}^{2}(3x^2/4-1)dx=[x^3/4-x]_{2/\sqrt{3}}^{2}$。

M1	正确新积分界及减基准厚度
M1	正确积分
A1	精确面积和单位

Question 15

P2-C08-R15 · 8 marks

(a) · 3 marks

$y = 5x/3 + 10/3$.

两个端点为(1,5)、(4,10)，斜率5/3，代点确定截距10/3。

M1	正确计算两端点
M1	斜率与点斜式
A1	方程

(b) · 3 marks

$102/5$.

先写成x^(3/2)+4x^(−1/2)，原函数2x^(5/2)/5+8√x，代界得62/5+8=102/5。

M1	化成可积分幂次
M1	正确原函数
A1	精确定积分

(c) · 2 marks

$21/10$ square units.

弦下是宽3、两端高度5和10的梯形，面积45/2；减曲线下面积102/5，得21/10。

M1	正确求弦下梯形面积或积分
A1	正面积差

Question 16

P2-C08-R16 · 7 marks

(a) · 3 marks

10.5 square units.

节点0、1、2、3的高度1、2、4、8，步长1，估计½[1+8+2(2+4)]=10.5。本单元不需对2^x作解析积分。

B1	四个正确节点高度
M1	梯形公式
A1	估计

(b) · 2 marks

13.5 square units.

矩形面积24，减下方估计10.5，得到13.5。

M1	矩形减曲线下方估计
A1	结果

(c) · 2 marks

Underestimates: the below-curve area was overestimated, so its complement was underestimated.

梯形在曲线上方使所扣面积偏大，固定矩形减去偏大量就得到偏小补集。

B1	指出曲线下面积的误差方向
B1	正确推出补集误差方向

Question 17

P2-C08-R17 · 9 marks

(a) · 3 marks

$20/3$ square units.

左段积分8/3，右段为底2高4的三角形面积4，总20/3。接点x=0两段均高4。

M1	分别处理两段定义
A1	两部分面积
A1	总面积

(b) · 3 marks

$8/3 + 4a - a^2 = 10/3 \Rightarrow a^2 - 4a + 2/3 = 0.$

左半目标是总面积的一半10/3，必须包括x<0的全部8/3以及0到a的直线下面积。

M1	设定一半总面积
M1	包括已存在的左曲线区域
A1	推出二次式

(c) · 3 marks

$a = 2 - \sqrt{30}/3.$

两根2±√(10/3)；正号大于2违背题设，故选负号。

M1	解二次式
A1	精确有效根
B1	明确排除另一根的范围理由

Question 18

P2-C08-R18 · 7 marks

(a) · 2 marks

$(-1, 1), (3, 9)$.

联立$x^2-2x-3=0=(x+1)(x-3)$，代回$y=x^2$。

M1	联立并解二次式
A1	两点

(b) · 3 marks

$5/3$ square units.

左区在-1至0，直线在上，积分 $\int_{-1}^{0}(2x+3-x^2)dx=[x^2+3x-x^3/3]_{-1}^{0}=5/3$。

M1	正确上下函数与界
M1	积分
A1	精确面积

(c) · 2 marks

$5:27$.

右区积分0到3，原函数在3为9，因此$R=9$；$L:R=(5/3):9=5:27$。横轴左右宽度之比不是面积比。

M1	求右区面积
A1	正确最简比

Question 19

P2-C08-R19 · 7 marks

(a) · 2 marks

4 and -3, respectively.

左区有向贡献+7，右区−3；整段积分7−3=4，右段为−3。

M1	正确赋予有向面积符号
A1	两个积分

(b) · 3 marks

2.

y=f(2x)在水平方向压缩至原来的一半，两区有向面积均减半：7/2−3/2=2；所给界正好是原横坐标的一半。

M1	正确横向压缩倍数
M1	同步映射区间
A1	积分值

(c) · 2 marks

4.

y=f(−x)是纵轴反射，正区面积仍7、负区仍3，只是交换左右位置；积分界变为−4到2，故总积分仍4。

M1	纵轴反射及新区间
A1	保留正负面积并得到4

Question 20

P2-C08-R20 · 7 marks

(a) · 2 marks

The consecutive widths are $1, 2, 3$ m, not equal.

梯形法同一步长公式要求横坐标等间隔；此表的三段间隔不同，必须逐个梯形计算。

B1	三个间隔正确
B1	指出等间隔条件不满足

(b) · 3 marks

$18\ \text{m}^2$.

三个梯形分别½×1×(1+2)=1.5、½×2×(2+4)=6、½×3×(4+3)=10.5，相加18。

M1	每段使用自己的宽度
A1	三面积计算
A1	合计与单位

(c) · 2 marks

3 tins.

两面用量2×18×0.4=14.4L；14.4/5=2.88，需要向上取整为3，不是按最近整数估算面积。

M1	同时计两面和每面涂覆率
A1	最少整罐数

Question 21

P2-C08-R21 · 9 marks

(a) · 3 marks

$[x^3/3 - kx^2 + kx]_0^1 = 1/3.$

逐项积分，上限处−k与+k相消；不能据图形高低直觉断言每个参数都改变面积。

M1	正确含参数原函数
M1	代界并消去参数
A1	不变量

(b) · 3 marks

$0 \le k \le 1.$

端点非负给k⩾0与1−k⩾0，是必要条件；这时 $f_k = (x-k)^2 + k(1-k) \ge 0$，所以也是充分条件。

M1	两端点给必要范围
M1	配方证明全段非负
A1	完整充分必要范围

(c) · 3 marks

Estimate $1/2$; error $1/6$.

两端高度k与1−k之和恒1，宽1，所以梯形面积½；真积分1/3，绝对误差1/6。两者均由前述不变量给出。

M1	用端点高度和求梯形
A1	估计及误差
B1	独立于k的解释

Question 22

P2-C08-R22 · 9 marks

(a) · 4 marks

$l: y = 2x - 4$; $C - l = x^2 - 6x + 9 = (x-3)^2$.

求导2x−4，x=3时斜率2，点为(3,2)，得切线。原函数减切线后配方。

M1	求切线
A1	切线正确
M1	两函数相减
A1	完全平方关系

(b) · 3 marks

$16/3$ square units.

差函数全段非负，可直接积分 $\int_1^5 (x-3)^2 dx$，或展开为x^2−6x+9；结果16/3。

M1	正确积分界与差函数
M1	正确积分代界
A1	精确面积

(c) · 2 marks

The separation $(x-3)^2$ is symmetric about $x = 3$, and the two vertical boundaries are equally distant from it.

两条原边界不必各自关于x=3对称；只需垂直差函数对称且左右积分宽度相同，就能保证两块面积相等。

B1	使用差函数的对称性
B1	同时说明界限等距

Question 23

P2-C08-R23 · 7 marks

(a) · 2 marks

6 square units.

区间可加性给 $\int_2^5 f = -1 - 5 = -6$，函数非正，所以几何面积为6。

M1	用积分可加性
A1	由非正函数转成正面积

(b) · 3 marks

$k = 2$.

移高k在宽3区间增加有向积分3k，因此−6+3k=0，得k=2。

M1	常数移高产生3k
A1	正确积分方程
A1	k

(c) · 2 marks

No. Its signed integral is zero, but the sizes of its positive and negative regions are not individually known.

仅知道正负有向面积相等，没有各自数值；移高后过轴位置也未知，因此无法确定总几何面积。

B1	不能确定
B1	说明缺少正负分区或面积信息

Question 24

P2-C08-R24 · 9 marks

(a) · 4 marks

$45/2 - 3\sqrt{3} - 3\sqrt{6} \approx 9.96\ \text{m}^2$.

先估曲线下方面积，步长3，估计3/2[1+4+2(1+√3+1+√6)]；从宽9、高4的矩形36中扣除。

M1	正确节点高度与步长
M1	梯形下方面积
M1	矩形补集
A1	估计面积

(b) · 3 marks

$9\ \text{m}^2$.

曲线与y=4在x=9相交，面积 $\int_0^9 (3 - \sqrt{x})dx = 27 - 18 = 9$。

M1	正确差函数及界
M1	正确积分
A1	精确面积

(c) · 2 marks

Not viable: $9 < 9.5$, although the estimate is above 9.5.

估计约9.96会误判达标；它是上方补集的高估，准确积分9低于规定门槛。

B1	正确决定
B1	估计可能误导的数值依据

Question 25

P2-C08-R25 · 9 marks

(a) · 3 marks

$k = 4$.

两面积分别 $\int_0^4 x^2 dx = 64/3$ 与 $k\int_0^4 \sqrt{x} dx = 16k/3$，相等给4。

M1	正确列两个积分面积
A1	两积分正确
A1	k

(b) · 2 marks

$a = 4^{2/3}$.

解$x^2=4\sqrt{x}$，除$\sqrt{x}$只在x>0合法，得x^(3/2)=4，故a=4^(2/3)；原点也是交点但此问只要正点。

M1	在正数范围化幂方程
A1	正交点横坐标

(c) · 4 marks

$32/3$ square units.

在0至a根式曲线在上，此块面积 $\int_0^a (4\sqrt{x} - x^2) dx = (8/3)a^{3/2} - a^3/3 = 16/3$。两原总面积相等使两差区域同面积，所以总为32/3。

M1	正确第一分区积分
M1	使用a^(3/2)=4与a^3=16
A1	单块16/3
B1	利用原总面积相等得到另一块同大

Question 26

P2-C08-R26 · 8 marks

(a) · 3 marks

$x^2 - 4x + h = 0 \Rightarrow (x-2)^2 = 4 - h.$

将曲线与水平线联立并配方。

M1	联立
M1	配方
A1	两根

(b) Given that the chord has length 2, find

(i) · 2 marks

$h = 3.$

两横坐标差为$2\sqrt{(4-h)}=2$，得$h=3$。

M1	弦长关系
A1	h

(ii) · 3 marks

$4/3$ square units.

此时交点1、3，积分$\int_1^3[x(4-x)-3]dx$，原函数$-x^3/3+2x^2-3x$，两端代值相减。

M1	正确差与积分界
M1	原函数
A1	精确面积

Question 27

P2-C08-R27 · 6 marks

(a) · 2 marks

13.0.

步长1，$T=\frac{1}{2}[2+4+2(2.8+3.4+3.8)]=13$。

M1	步长与端点权重
A1	估计值

(b) For $0 \le x \le 4$, the upper curve is $y=2f(x)+3$ and the lower line is $y=1$.

(i) · 2 marks

34.0.

竖直间距$2f(x)+2$，面积$2I+8$，用$I \approx 13$得34；不用重新列表。

M1	面积转为2I+8
A1	估计值

(ii) · 2 marks

Underestimate.

原曲线高于各弦，梯形和低估I。对面积$2I+8$，乘以正数2并加常数不改变误差方向，所以仍为低估。

B1	原梯形误差方向
B1	变换后仍低估

Question 28

P2-C08-R28 · 6 marks

(a) · 2 marks

$46:29$.

下方面积估计18.4，上方面积$30-18.4=11.6$；比$184:116=46:29$。

M1	用矩形减积分
A1	最简比

(b) The curve lies below each straight chord used by the trapezium rule.

(i) · 2 marks

Too small.

梯形高估曲线下方面积，而上方面积是固定矩形减去该值，所以低估上方面积。

B1	下方面积高估
B1	余面积低估

(ii) · 2 marks

Too large.

分子被高估、正的分母被低估，两种影响都使下方与上方之比偏大。

B1	识别分子与分母的误差
B1	比值方向

Question 29

P2-C08-R29 · 8 marks

(a) · 3 marks

$l: y=-2x+5$; separation $(x-1)^2$.

$y\prime=2x-4$，切点$(1,3)$，切线$y=-2x+5$；相减x^2-2x+1。

M1	切线方程
A1	切线
A1	平方差距

(b) Find

(i) · 2 marks

3.

$\int_0^3(x-1)^2dx=[(x-1)^3/3]_0^3=3$。

M1	非负差积分
A1	面积

(ii) · 3 marks

$a=1+\sqrt[3]{7/2}$.

左部面积$[(a-1)^3+1]/3=3/2$，得$(a-1)^3=7/2$；实根在所要求区间。

M1	等面积方程
M1	正确积分上下限
A1	参数

Question 30

P2-C08-R30 · 7 marks

(a) · 3 marks

$f(x) = 2x^2 - 6x + 4$; $(2, 0)$.

积分得$2x^2 - 6x + c$，代入$(1, 0)$得$c = 4$，所以$f(x) = 2(x - 1)(x - 2)$。

M1	重建曲线
A1	常数与因式
A1	另一个截距

(b) Let A be the total area between the curve and the x-axis on $0 \le x \le 2$.

(i) · 2 marks

2.

在$[0, 1]$函数非负、在$[1, 2]$非正。原函数$2x^3/3 - 3x^2 + 4x$在$1, 2$分别为$5/3, 4/3$；总面积$5/3 + (5/3 - 4/3) = 2$。

M1	在零点分段取正面积
A1	总面积

(ii) · 2 marks

$4/3$.

有向积分$4/3$，将轴下方的$1/3$作为负数；几何总面积则将其作为正数，所以与2不同。

M1	有向积分
A1	正确解释差异